

FH Aachen

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Expose

Evaluating AI Agent Capabilities in SysML v2 Modeling via a Semantic MCP Interface

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- [2] OBJECT MANAGEMENT GROUP: Kernel Modeling Language (KerML), Version 1.0 / Object Management Group. Version: März 2026. <https://www.omg.org/spec/KerML/1.0/PDF>. 2026 (formal/26-03-01). – Specification. – Normative OMG specification of the logic-grounded kernel SysML v2 extends, with four declarable tool conformance levels. Supplies the vocabulary for what an MCP bridge exposes – structural CRUD versus semantic conformance – and why v2 constraints are natively verifiable.
- [3] BOELSEN, Kathrin ; MAY, Matthias ; JACOBS, Georg ; MENNICKEN, Manuel ; MOERS, Frederik ; ZERWAS, Thilo ; HÖPFNER, Gregor: SysML v2 based modelling guidelines for mechanical system elements. In: *Forschung im Ingenieurwesen* 89 (2025), No. 1, p. 60. <http://dx.doi.org/10.1007/s10010-025-00827-w>. – DOI 10.1007/s10010-025-00827-w. – Encodes mechanical-domain modelling guidelines natively in SysML v2 via Metadata Definitions (moteago methodology, one worked example, no quantitative evaluation). Precedent for exposing guideline conformance as a tool-native check in *Validierung*.
- [4] AHLBRECHT, Alexander ; ZAESKE, Wanja ; DURAK, Umut: MBSqlE: Enabling SQL as Powerful Query Language for SysML v2 in Aviation. In: *44th AIAA DATC/IEEE Digital Avionics Systems Conference (DASC 2025)*. Montreal, Canada : IEEE, September 2025. – The standard SysML v2 API offers no graph traversal (REST traversals >10 s); MBSqlE imports models into SQLite so recursive SQL traverses them in one query. Motivates a semantic/cache layer instead of a thin REST proxy for *Abfragen*.
- [5] HELLE, Philipp ; SCHRAMM, Gerrit: Fly me to the Moon – Modeling Apollo 11 Using SysML v2. In: *Systems Engineering* (2026), Juni. <http://dx.doi.org/10.1002/sys.70074>. – DOI 10.1002/sys.70074. – [TOP 3] Airbus Central R&T’s five-layer CoSMA (Purpose/Operational/Functional/Logical/Technical) SysML v2 model of the Apollo 11 mission, released open-source (github.com/airbus/apollo-11-sysml-v2, MPL-2.0, commit 6e9c93f, accessed 20.09.2026) as a public MBSE benchmark. The instrument this thesis mea-

sures on: ~7.2 kLOC / 28 files / ~2 000 elements, deliberately incomplete "scaffold" for *Erstellen/Korrektur* tasks.

- [6] GESELLSCHAFT FÜR SYSTEMS ENGINEERING (GFSE): *SysML-v2-Models: An Open Collection of SysML v2 Models*. GitHub repository, <https://github.com/GfSE/SysML-v2-Models>, commit ebbb0c3, accessed 20.09.2026, 2025. – BSD-3. ~5.1 kLOC curated teaching/reference models (VehicleModel, DroneModelLogical, EveOnline Mining Frigate) explicitly meant to "help train LLMs"; ships `CI/parse.py` + a pilot-implementation jar as a validity oracle. Secondary/generalisation set (D2) – H3 target, small self-contained models plus deliberate "bad examples".
- [7] FRESEMANN, Carina ; ELLSEL, Claudius ; GRUNENWALD, Carl-Philipp ; STARK, Rainer: Using Large Language Models for System Modeling: Current State and Future Implications. In: *2025 IEEE International Symposium on Systems Engineering (ISSE)*, IEEE, Oktober 2025, p. 1–8. – Structured review of 20 publications on LLMs for MBSE tasks (creation, analysis, modification, reformulation). Supplies the taxonomy behind this thesis's use-case catalogue, names MCP as the emerging context mechanism, and flags the missing evaluation criteria.
- [8] DEHART, John K.: Leveraging Large Language Models for Direct Interaction with SysML v2. In: *INCOSE International Symposium 34 (2024)*, Juli, No. 1, p. 2168–2185. <http://dx.doi.org/10.1002/iis2.13262>. – DOI 10.1002/iis2.13262. – Founding paper of the LLM×SysML v2 topic: conceptual analysis and case studies, no controlled evaluation or ground truth. Provides the vision the thesis tests empirically and the "LLM edits text directly" pattern behind the no-tool arm.
- [9] DEHN, Simon ; SCHNÜRER, Simon ; JACOBS, Georg ; HÖPFNER, Gregor: Generating SysML v2 Models from Natural Language Requirements Using Large Language Models. In: *2025 IEEE International Symposium on Systems Engineering (ISSE)*, IEEE, Oktober 2025, p. 1–7. – Four-component structured prompting for NL→SysML v2 generation, evaluated on an automotive use case with precision/recall/F1 and syntax checks over three runs; few-shot examples prove essential. Primary reference for *Erstellen*, but file-based without a modelling tool.
- [10] QUALIS, Richard A.: Mitigating Hallucinations in SysML v2 Generation Using LLMs and a Tri-Layered Knowledge Graph Reasoning Framework. In: *2025 ACM/IEEE 28th International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, IEEE, Oktober 2025, p. 357–366. – Tri-layered knowledge graph grounding SysML v2 generation against hallucinations, evaluated on a satellite case study against curated ground truth: structurally valid, semantically weak. Motivates grounding the agent in the live model rather than a static KG (*Erstellen*).

- [11] WU, Rui ; WEN, Guangwei ; ZHANG, XiaoNing ; LIANG, Xiaojun ; YANG, Chunhua ; GUI, Weihua: Knowledge-Augmented Conversational Agents for Dependency-Aware SysML v2 Model Generation and Refinement. In: *2025 IEEE 26th China Conference on System Simulation Technology and its Applications (CCSSTA)*, IEEE, Juli 2025, p. 581–586. – SysForge: four agents with dependency-aware KG retrieval and a generate–validate–refine loop, beating plain-LLM and semantic-RAG baselines on Pass@1 and BLEU. Closest prior work for *Korrektur*, but file-based with its own validator instead of tool-native validation.
- [12] QUAST, Vincent ; JACOBS, Georg ; DEHN, Simon ; HÖPFNER, Gregor: Enabling Humans and AI Systems to Retrieve Information from System Architectures in Model-Based Systems Engineering. In: *Systems* 14 (2026), Januar, No. 1, p. 83. <http://dx.doi.org/10.3390/systems14010083>. – DOI 10.3390/systems14010083. – GraphRAG multi-agent QA over a Neo4j graph parsed from SysML v2 text, reaching 93 % accuracy on a synthetic battery-EV model. Accuracy-metric template for *Abfragen* and a second “no standardized benchmarks” gap citation; it queries a derived graph, not the live model.
- [13] BOUAMRA, Yasmine ; YUN, Bruno ; POISSON, Alexandre ; ARMETTA, Frédéric: SysTemp: A Multi-Agent System for Template-Based Generation of SysML v2. In: *Advances in Practical Applications of Agents, Multi-Agent Systems, and Computational Social Science: The PAAMS Collection – 23rd International Conference, PAAMS 2025, Lille, France* Bd. 16031, Springer Nature Switzerland, Juni 2025 (Lecture Notes in Computer Science (LNAI)), p. 41–52. – Multi-agent template pipeline for NL-to-SysML v2 generation; scaffolding reaches zero syntax errors in 4/5 toy scenarios vs. 1/5 without, syntax only. The gap citation for this thesis’s benchmark (“absence of standard benchmarks”) and for SysML v2 corpus scarcity.
- [14] AL-SHAMI, Haitham ; MALIK, Rohail ; ALA-LAURINHO, Riku ; VEPSÄLÄINEN, Jari ; VIITALA, Raine: Automated Semantic Fault Localization in SysML v2: A Human-in-the-Loop Framework Using Knowledge-Graph Augmented LLMs. In: *INCOSE International Symposium* 36 (2026), Juli, No. 1, p. 1274–1286. <http://dx.doi.org/10.1002/iis2.70189>. – DOI 10.1002/iis2.70189. – Fine-tuned small LLM plus a physical-compatibility KG repairs *semantic* faults that pass SysML v2 syntax checks, rising from <3 % to >91 % on 1 184 test samples. Methodological precedent for the fault injection and fault taxonomy of *Korrektur*.
- [15] BAZZAL, Mahmoud ; SCHLEIFER, Simon ; LUNGU, Adriana ; KRUSE, Benjamin ; GOETZ, Stefan ; KOLDEWEY, Christian ; TISSEN, Denis ; WARTZACK, Sandro J. ; DUMITRESCU, Roman: Model Context Protocol for MBSE: A Unified Interoperable Agentic Framework for Systems Modelling. In: *Proceedings of the Design Society* 6

- (2026), Juli, p. 2721–2730. <http://dx.doi.org/10.1017/pds.2026.10630>. – DOI 10.1017/pds.2026.10630. – [TOP 3] MCP servers on the standard SysML v2 API with creation, modification and analysis tools, shown only through two qualitative case studies without metrics. Closest competing work: it delivers the artifact class, the thesis the missing capability evaluation inside MSoSA.
- [16] DEMPSEY, Colin: *Native AI Integration for Model-Based Systems Engineering: Three Layers that Make It Work*. SEI Blog, Carnegie Mellon University, <https://www.sei.cmu.edu/blog/native-ai-integration-for-model-based-systems-engineering-three-layers-September-2026>. – [TOP 3] Controlled three-arm SysML v2 experiment (no tool / validation loop / plus MCP knowledge and skills): both tool arms reach zero syntax errors, but only full tooling lifts the pattern score (71.7 → 94.1). Direct template for this thesis’s ablation and evidence for H2.
- [17] ISO/IEC/IEEE: Systems and software engineering – System life cycle processes / International Organization for Standardization. Version: Mai 2023. <https://www.iso.org/standard/81702.html>. 2023 (ISO/IEC/IEEE 15288:2023). – International Standard. – Defines Verification (conformance to specified requirements) and Validation (fulfilment of stakeholder needs in the operational environment) as distinct life-cycle processes. The thesis adopts these definitions for its *Verifizierung* and *Validierung* use cases and audits prior work against them.
- [18] CIBRIÁN, Eduardo ; OLIVERT-ISERTE, Jose ; DÍEZ-FENOY, Carlos ; MENDIETA, Roy ; LLORENS, Juan ; ÁLVAREZ-RODRÍGUEZ, Jose M.: Ensuring Semantic Consistency in SysML v2 Models Through Metamodel-Driven Validation. In: *IEEE Access* 13 (2025), p. 121444–121457. <http://dx.doi.org/10.1109/ACCESS.2025.3587786>. – DOI 10.1109/ACCESS.2025.3587786. – Metamodel-derived rule checking (lexical, ANTLR syntax, semantic cross-references) detects 97 % of 428 injected errors across 25 OMG test models. Precedent for *Verifizierung*: an error taxonomy, a fault-injection method transferable to *Korrektur*, and a detection-rate baseline.
- [19] MOLNÁR, Vince ; GRAICS, Bence ; VÖRÖS, András ; TONETTA, Stefano ; CRISTOFOROTTI, Luca ; KIMBERLY, Greg ; DYER, Pamela ; GIAMMARCO, Kristin ; KOETHE, Manfred ; HESTER, John ; SMITH, Jamie ; GRIMM, Christoph: Towards the Formal Verification of SysML v2 Models. In: *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems (MODELS Companion '24)*, 2024, p. 1086–1095. – OMG Formal Methods WG survey of five formal-methods integrations for SysML v2 (model checking, theorem proving, contracts), illustrated by small examples without benchmarks. State-of-the-art backdrop for positioning a prompt-driven agent’s *Verifizierung* capability, which has no formal backend.

- [20] RATZKE, Axel ; KOCH, Johannes ; GRIMM, Christoph: Modeling and Analysis of System Models with Constraints in SysMLv2. In: *2025 20th Annual System of Systems Engineering Conference (SoSE)*, 2025, p. 1–6. – Extends SysML v2 constraints with range semantics (`oneOf/anyOf/allOf`) compiled into a CSP, demonstrated on two small examples and limited to static structure. Mechanism reference for what an agent could delegate to a tool-native constraint solver in *Verifizierung*.
- [21] LIPARULO, Leonardo ; PIERRI, Francesco: *Benchmarking AI Agents for Hardware Design Automation via MCP Tool Calling*. arXiv:2608.26199 [cs.AI]. – Benchmarks seven open-source models against a purpose-built MCP server for hardware design across single-operation, chained, invalid and misspelled tasks under prompt and tool-description ablations. Closest methodological sibling: its task tiers and tool-description findings shape the bridge and benchmark design.
- [22] HASAN, Mohammed M. ; LI, Hao ; RAJBAHADUR, Gopi K. ; ADAMS, Bram ; HASSAN, Ahmed E.: *Model Context Protocol (MCP) Tool Descriptions Are Smelly! Towards Improving AI Agent Efficiency with Augmented MCP Tool Descriptions*. arXiv:2602.14878. – Study of 856 MCP tools: 97.1 % of descriptions carry a “smell”, and augmenting them raises success by 5.85 pp but adds 67 % more steps. Supplies the rubric for the bridge’s tool descriptions and a confound to hold constant across arms.
- [23] MOLINARI, Gioele ; FELTEN, Florian ; MASSOUDI, Soheyl ; FUGE, Mark: *EngiAI: Capability-Based Evaluation of Tool-Connected LLM Agents for Engineering Design*. arXiv:2605.19743 (extended version of an ASME IDETC-CIE 2026 paper). – Capability-based evaluation scoring workflow execution, parameter selection, orchestration and code authoring separately from execution traces; backends reach 96–97 % on workflow tasks but only 20–53 % on tool-based decisions. Rationale for per-use-case scoring and template for the failure taxonomy.
- [24] PRADAS-GOMEZ, Alejandro ; BRAHMA, Arindam ; ISAKSSON, Ola: *DUCTILE: Agentic LLM Orchestration of Engineering Analysis in Product Development Practice*. arXiv:2603.10249. – Splits adaptive orchestration by the agent from deterministic execution by verified tools, evaluated on an industrial aerospace task against expert acceptance criteria over 10 repeated runs. Template for the agent–tool division of labour and for repeated-run reliability reporting.
- [25] WANG, Zizhe: *Beyond Executable Models: The Pufibara Agent Harness and the Modelica Agent Workflow Benchmark for Physical System Modeling*. arXiv:2608.23653 [cs.SE]. – Agent harness plus a source-grounded 232-task Modelica benchmark (repair, generation, tuning) scored by an evaluator outside the agent loop. Supplies three

adopted principles: external evaluator, tasks grounded in an existing model, and the repair/generation/tuning split.

- [26] SHI, Jiasheng ; ZHANG, Tianhan: *What Do CAE Simulation Agents Really Need Beyond a Generic Harness?* arXiv:2609.03718 [cs.CE]. – With information access and repair budget held fixed, a generic single-agent harness beats specialised multi-agent CAE systems (96.4 % vs. 88.2 %); execution-feedback repair and domain tutorials drive the gains. Counter-evidence to H2 that the thin-vs-semantic ablation must engage.
- [27] SHEFA, Jannatul ; SALADO, Alejandro ; WACH, Paul ; TOPCU, Taylan G.: *Two Truths and A Lie? Benchmarking Off-the-Shelf LLMs for Requirements Quality Assessment: Performance, False Alarms, and Misses.* arXiv:2609.03230 [cs.SE]. – Ten LLMs, 100 runs, expert ground truth on INCOSE criteria: the best model finds a median of 47 % of issues and misses judgement-heavy ones. Justifies expert acceptance criteria with a human second rater instead of LLM-as-judge, and repeated runs.